

Python: an introduction

- program structure: constants, acquisition, computation, visualisation
- variables and constants
- `input()` and `print()`
- `import` statement to access existing python libraries, for instance `import math:` to be put at the beginning of our program

Details

- `input()` always returns a string, a sequence of character, therefore if that data represents a value (integer or real), it needs to be converted, by means of `int()` or `float()`.
- given a value (integer or real) it is possible to covert and obtain a string consisting of the numerical characters (and the "." separator) representing the value into a string, by means of the `str()` function.

Exercises in class

Esercizio svolto #1:

Write a Python program that receives in input a positive integer value (it is surely so) representing the side of a square and computes the area. Finally, it shows the result.

```
1 # CONSTANT / ARBITRARY ASPECTS OF THE SOLUTION
2
3 # DATA ACQUISITION
4 side = int(input())
5
6 # COMPUTATION
7 area = side * side
8
9 # VISUALISATION
10 print(area)
```

Esercizio svolto #2:

Write a Python program that receives in input a positive integer value (it is surely so) representing the radius of a circle and it computes the length of the circumference and its area. Finally, it shows the results.

Initial version:

```
1 # CONSTANT / ARBITRARY ASPECTS OF THE SOLUTION
2 PI = 3.14159    /* it is a constant value: let us put a meaningful label and use it */
3
4 # DATA ACQUISITION
5 r = int(input())
6
7 # COMPUTATION
8 circ = 2 * r * PI
9 area = r * r * PI
10
11 # VISUALISATION
12 print(circ, area)
```

Note: for established values it is possible to think that there is a library available. This is the case of the `math` library.

```
1 # CONSTANT / ARBITRARY ASPECTS OF THE SOLUTION
2 import math
```

```

3
4 # DATA ACQUISITION
5 r = int(input())
6
7 # COMPUTATION
8 circ = 2 * r * math.pi
9 area = r * r * math.pi
10
11 # VISUALISATION
12 print(circ, area)

```

Note: avoid putting constants directly in the code. It reduces readability and flexibility.

Esercizio svolto #3:

Write a python program that asks for a positive integer, representing a time expressed in seconds; the program converts it into days, hours, minutes, and seconds, and finally shows the result.

```

1 # CONSTANT / ARBITRARY ASPECTS OF THE SOLUTION
2 SEC_IN_MIN = 60
3 MIN_IN_HR = 60
4 HR_IN_DAY = 24
5 SEC_IN_HR = (SEC_IN_MIN * MIN_IN_HR)
6 SEC_IN_DAY = SEC_IN_HR * HR_IN_DAY
7
8 # DATA ACQUISITION
9 timesec = int(input())
10
11 # COMPUTATION
12 num_days = timesec // SEC_IN_DAY
13 trem = timesec % SEC_IN_DAY
14
15 num_hr = trem // SEC_IN_HR
16 trem = trem % SEC_IN_HR
17
18 num_min = trem // SEC_IN_MIN
19 num_sec = trem % SEC_IN_MIN
20
21 # VISUALISATION
22 print(num_days, num_hr, num_min, num_sec)

```

Proposed exercises

Esercizio proposto #1:

Write a python program that, after receiving an integer value, calculates and displays 1 if the value is positive, 0 otherwise.

Esercizio proposto #2:

Write a python program that, after receiving an integer value, calculates and displays 1 if the value is odd, or 0 otherwise. Make a variation that does not use the remainder (modulus) operator.

Esercizio proposto #3:

Write a python program that takes three positive values and determines whether they form a Pythagorean triple. If they do, display 1; otherwise, display 0.

Esercizio proposto #4:

Write a python program that takes an integer value and displays + if the value is positive, - if it is negative, and a space character if it is zero.

Esercizio proposto #5:

Write a python program that takes an integer value in the range [1, 12], representing a month of the year (1 = January, etc.) and computes and displays the number of days in that month.

Esercizio proposto #6:

Write a python program that takes two integer values: the first is the month, expected to be in the interval [1, 12], the second one is the year: the program computes the number of days in that month, taking into account also the fact the year is a leap year or not.

Esercizio proposto #7:

Create a program to split a restaurant bill. The program receives in input the total bill amount and the number of people sharing it. Apply a fixed tip percentage constant, then output the total tip, the total amount due, and each person's share.

Esercizio proposto #8:

Write a program that receives in input a real value representing a temperature expressed in Celsius and computes and visualizes the same temperature expressed in Fahrenheit. The formula for the relationship is $c = (f - 32) * 5/9$.

Esercizio proposto #9:

Write a program that receives in input a real value representing a temperature expressed in Fahrenheit and computes and visualizes the same temperature expressed in Celsius.

Esercizio proposto #10:

Write a program that receives an integer value in input and a real one corresponding to height (in cm) and weight (in kg), respectively. The program computes and shows the BMI of the person. The BMI is calculated by dividing the weight in kilograms by their height in metres squared.

Esercizio proposto #11:

Write a program that receives in input the height of a person expressed in centimeters and computes the ideal weight according to the BMI formula (use $BMI = 22$)